

Kendriya Vidyalaya Sangathan, Varanasi Region

Pre- Board - I (2024 – 25)

Class : X

Max. Marks : 80

Subject : Mathematics (Standard) (041)

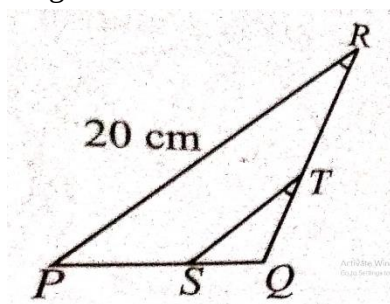
Max. Time : 3 Hours

General Instructions:

1. This question paper has five sections A, B, C, D and E
2. **Section A** has 20 Multiple Choice Questions (MCQs) carrying 1 mark each.
3. **Section B** has 5 Short Answer- I (SA-I) type questions carrying 2 marks each.
4. **Section C** has 6 Short Answer - II (SA -II) type questions carrying 3 marks each.
5. **Section D** has 4 long Answer (LA) type questions carrying 5 marks each.
6. **Section E** has 3 source based/ case based/passage based/ integrated units of assessment (4 marks each) with sub parts of the values of 1, 1 and 2 marks each respectively.
7. All questions are compulsory. However, an internal choice in 2 questions of 2 marks, 2 questions of 3 marks and 2 questions of 5 marks has been provided. An internal choice has been provided in the 2 marks questions of Section E.
8. Draw neat figures wherever required. Take $\pi = 22/7$ wherever required if not stated.

Section - A

1. If the $\text{HCF}(2520, 6600) = 40$ and $\text{LCM}(2520, 6600) = 252 \times k$, then the value of k is :
(a) 1650 (b) 1600 (c) 165 (d) 1625
2. The value of k for which the system of equations $8x - 4y + 10 = 0$; $kx - y + 5/2 = 0$ are coincident is:
(a) -2 (b) 2 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$
3. $\triangle PQR$ is shown below. ST is drawn such that $\angle PQR = \angle STQ$. If ST divides QR in the ratio $2 : 3$ and $PR = 20$ cm, then what is the length of ST ?



- (a) $\frac{10}{3}$ cm (b) 8 cm (c) 12 cm (d) $\frac{40}{3}$ cm
4. The quadratic equation : $ax^2 + bx + c = 0$ has real and equal roots then c is equal to :
(a) $\frac{-b}{2a}$ (b) $\frac{b}{2a}$ (c) $\frac{b^2}{4a}$ (d) $\frac{-b^2}{4a}$
 5. The ratio of arc lengths of minor sector and major sector is $2 : 7$. The angle made by minor arc at the centre of the circle is :
(a) 80° (b) 60° (c) 90° (d) 120°
 6. If $C(-2, 3)$ is the centre of the circle having the end points $A(4, -2)$ and $B(x, 8)$ of a diameter AB , then the value of x is:

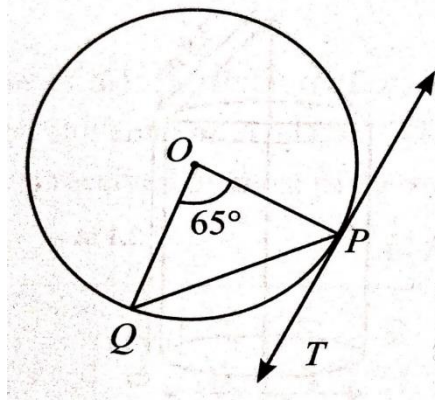
(a) 8

(b) 6

(c) - 6

(d) - 8

7. In the given figure, PT is tangent to the circle with centre O. Chord PQ subtends an angle 65° at the centre. The measure of $\angle QPT$ is:

(a) 65° (b) 57.5° (c) 22.5° (d) 32.5°

8. If n^{th} term of an A.P. is $7 - 4n$, then its common difference is;

(a) 4

(b) 3

(c) - 4

(d) - 3

9. Distance between the points $(m, -n)$ and $(-m, n)$ is :

(a) $\sqrt{m^2 + n^2}$ unit(b) $m + n$ unit(c) $2\sqrt{m^2 + n^2}$ unit(d) $\sqrt{2m^2 + 2n^2}$ unit

10. If d is the diameter of sphere, then the surface area of sphere is :

(a) $\frac{4}{3}\pi d^3$ (b) πd^2 (c) $4\pi d^2$ (d) $\frac{2}{3}\pi d^3$

11. If $a, a + d, a + 2d \dots$ is an A.P and S_n is the sum of first n terms, then :

(a) $S_2 - 2S_1 = d$ (b) $S_2 - S_1 = d$ (c) $2S_2 - S_1 = d$ (d) $S_3 - 2S_2 = d$

12. If $\sin A = \frac{1}{\sqrt{2}}$, then the value of $\cot 2A$ is:

(a) 1

(b)) Not defined

(c) $\sqrt{3}$

(d) 0

13. The value of $\sec A - \tan A = \frac{1}{7}$, then $\frac{1+\sin A}{\cos A}$ is equal to :

(a) -1

(b) 7

(c) -7

(d) 1

14. The ratio of lengths of pole and its shadow is $\sqrt{3} : 1$, then the angle of elevation of sun is:

(a) 60° (b) 45° (c) 30° (d) 90°

15. A fair coin is tossed twice, then the probability of getting no tails is :

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$

(d) 1

16. All the aces from the well shuffled deck of 52 card are removed, then a card is drawn at random, the probability that it is a black queen is :

(a) $\frac{1}{12}$ (b) $\frac{3}{12}$ (c) $\frac{1}{24}$ (d) $\frac{3}{24}$

17. The sum of lower limit and upper limit of the median class is :

Class	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
Frequency	5	8	7	12	18

(a) 50

(b) 70

(c) 90

(d) 30

18. If α and β are the zeroes of the polynomial, $(x) = 4x^2 - 3x - 7$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is equal to :

(a) $\frac{7}{3}$ (b) $-\frac{7}{3}$ (c) $\frac{3}{7}$ (d) $-\frac{3}{7}$

DIRECTION : In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R) has the following four choices (a), (b), (c) and (d), only one of which is the correct answer. Mark the correct choice.

(a) Assertion (A) and Reason (R) are True; Assertion (A) is a correct explanation for Reason (R).

(b) Assertion (A) and Reason (R) are True; Assertion (A) is not a correct explanation for Reason (R)

(c) Assertion (A) is True, Reason (R) is False.

(d) Assertion (A) is False, Reason (R) is True.

19. **Assertion (A)** : The length of minute hand of a clock is 7 cm, then the area swept by minute hand in 5 minutes is $12\frac{5}{6} \text{ cm}^2$.

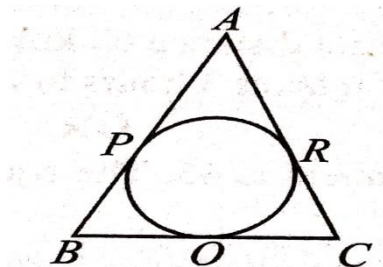
Reason (R) : The area of a sector of sector angle θ and radius r is given by $\frac{\theta}{360^\circ} \pi r^2 \text{ unit}^2$.

20. **Assertion (A)** : At a point P of a circle with centre O and radius 12 cm, a tangent PQ of length 16 cm drawn, then $OQ = 18 \text{ cm}$.

Reason (R) : The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Section – B

21. Find the largest number which exactly divides 70 and 125 leaving remainders 5 and 8 respectively.
 22. In the given figure, sides AB, BC and CA of a $\triangle ABC$, touch a circle at P, O and R respectively. If $AP = 5 \text{ cm}$, $BP = 4 \text{ cm}$. If the perimeter of triangle is 32 cm, then find the length of BC and AC.



23. In a $\triangle ABC$, $DE \parallel BC$, intersecting AB and AC at D and E respectively. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, then find the value of x . (All the measures are in cm)

OR

Using the converse of basic proportionality theorem prove that the line joining the mid – points of any two sides of triangle is parallel to the third side.

24. If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, $0^\circ < A + B \leq 90^\circ$ and $A > B$, then find the angles A and B.

25. The perimeter of a sector of a circle of radius 5.2 cm is 16.4 cm. Find the area of the sector

OR

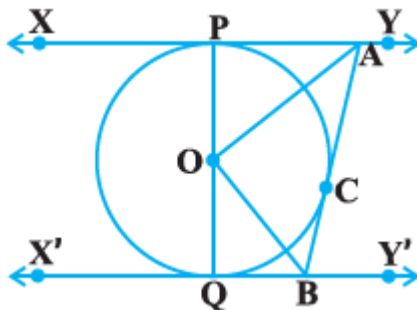
A chord of a circle of radius 7 cm subtends an angle 90° at the centre. Find the area of corresponding minor segment of the circle.

Section – C

26. Prove that $\sqrt{7}$ is an irrational number.

27. The age of Mohan is twice the sum of ages of his two children. After 20 years Mohan will be equal to the sum of ages of his two children. Find the age of Mohan.

28. In the given figure XY and X'Y' are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting XY at A and X'Y' at B. Prove that $\angle AOB = 90^\circ$.



OR

Two circles touch each other externally at P. AB is common tangent to the circles touching them at A and B. Find the value of $\angle APB$.

29. Find the modal weight of apples in the following frequency distribution.

Mass (in grams)	80 - 100	100 - 120	120 - 140	140 - 160	160 - 180
No. of apples	20	60	70	40	60

30. Prove that: $\frac{\tan A}{1+\sec A} - \frac{\tan A}{1-\sec A} = 2\operatorname{cosec} A$

OR

If $\sin\theta + \cos\theta = p$ and $\sec\theta + \operatorname{cosec}\theta = q$, prove that $q(p^2 - 1) = 2p$

31. Find the value of 'p' for which one root of a quadratic polynomial $px^2 - 14x + 8$ is 6 times the other.

Section – D

32. Solve: $\frac{x-2}{x-3} + \frac{x-4}{x-5} = \frac{10}{3}$, where $x \neq 3, 5$

OR

A train travels 180 km at a uniform speed. If the speed had been 9 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

33. State and prove Basic Proportionality Theorem.

34. The following frequency distribution shows the sales of athletic shoes at a online store each day during the past 100 days. If the median sale per day is 525 shoes, find the values of x and y.

Sales	No. of days
0 - 100	2
100 - 200	5
200 - 300	x
300 - 400	12
400 - 500	17
500 - 600	20
600 - 700	y
700 - 800	9
800 - 900	7
900 - 1000	4
Total	N = 100

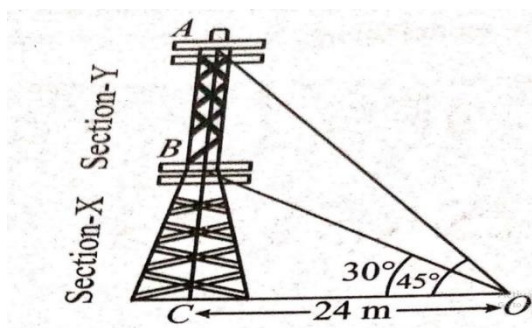
35. A welfare association ordered 100 tents from government for sudden flood in Bihar. The lower part of each tent is of the form of cylinder of diameter 4.2 m and height 4 m with the conical upper part of same diameter but of height 2.8 m. Find the total surface area of the 100 tents government supplied to the association. Also find the total cost of canvas used in making the tents at the rate of Rs 50 per m^2 .

OR

A vessel in the form of an inverted cone. Its height is 10 cm and the radius of its top is 5 cm. It is filled with water up to the brim. When lead shots, each of which is a sphere of radius 0.5 cm are dropped into the vessels, one fourth of water flows out. Find the number of lead shots in the vessel.

Section – E

36. Case Study-1



A radio station was build near village Mundera for the better transmission of signals and there were two sections, section – X and section – Y upon which tower was build. Engineers have observed that the angle of elevation of top of the section – X is 30° , when engineers were at a distance of 24 m from the base of radio station and angle of elevation of the top of section – Y is 45° . A wire is lied from top of radio station (from point – A) to a point O on the ground for the additional support. (Use $\sqrt{3} = 1.73$)

From above information, answer the following questions:

- Find the height of section – X.
- Find the total height of tower.
- Find the length of wire OA.

OR

Find the change in the height of the tower when the angle of elevation of the top of the tower A increases to 60° .

37. Case Study- 2

There are 810 apples which are arranged in such a way that 5 apples are in the first basket, 12 in the second basket, 19 in the third basket and so on..

From above information, answer the following questions :

- How many apples are there in the 9th basket?
- If in n^{th} basket 68 apples are kept, find the value of n.
- Find the total number of apples in first 13 baskets.

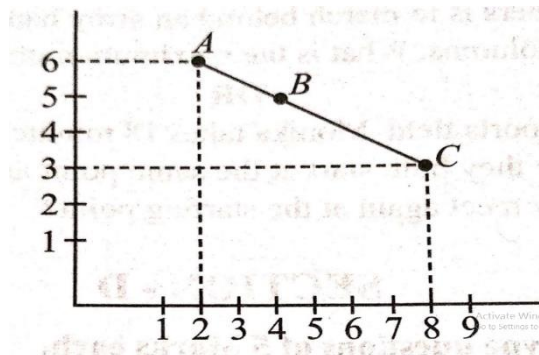
OR

In how many baskets are 810 apples kept?

38. Case Study – 3

Read the following text and answer the questions given below:

A group of class X students goes to picnic during summer holidays. Three friends Aman, Bhupesh and Chahal are sliding on a slider. The position of three friends are shown by the points A , B and C on the slider.



- Find the distance between A and C
- Is 'B' the mid – point of AC? Justify by finding the mid – point of AC.
- Find the ratio in which the point B divides the line segment AC.

OR

Find a point on x – axis which is equidistant from A and B.